ROOMAN
INNOVATE INTEGRATE EMPOWER

INTERNSHIP AT

ROOMAN TECHNOLOGIES

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COURSE STRUCTURE

ROOMAN
INNOVATE INTEGRATE EMPOWER

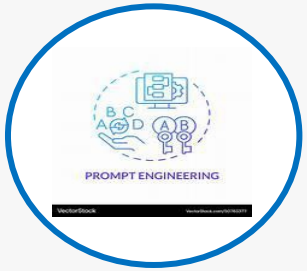
1. IT IGNITE
2. AI IN CYBER SECURITY
3. PROJECT

IT IGNITE

Week 1

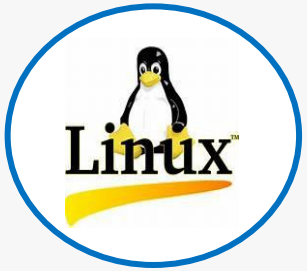
Python Programming

Week 2



Prompt Engineering

Week 3



Linux Basics

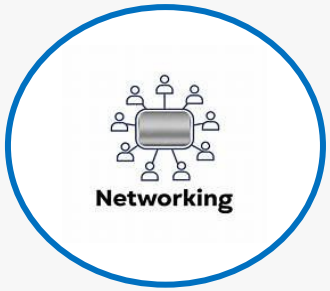
Week 4

AI Coding Assistants & Modern Dev Workflows

COURSE STRUCTURE

IT IGNITE

Week 5



Networking Essentials

Week 6

Cloud Fundamentals

Week 7

Cyber Security

Week 8

Database

Week 9

Software Engineering

Week 10

Version Control &Github

Week 11

Digital Literacy

Week 12

Workplace Readiness

WEEK-WISE TIMELINE

WEEK -1

- Orientation & Onboarding
- Program Induction
- LMS Introduction
- Programming Concepts
- Introduction to Python
- Google Colab

WEEK -2

- Introduction to Cybersecurity
- Information Security Principles
- Control Flow Statements
- History of Cybersecurity Attacks
- Virtual Cybersecurity Lab Setup

WEEK -3

- Python Essentials
- Data Structures
- Packet Capturing & Analysis
- Lambda Functions
- File Handling
- Offensive & Defensive Security,

WEEK -4

- Generative AI & LLMs
- Prompt Engineering
- Open Source Intelligence
- Advanced Prompting
- Ethical Prompting
- Nmap Scripts

WEEK -5

- Linux Basics
- Vulnerability Assessment
- Command Line Operations
- OWASP & ZAP
- DIRB Essentials

WEEK -6

- Linux in Cloud Environment
- Bash Scripting For Automation
- IDS, IPS, Honeypot
- Anonymity & Proxychain

TECHNICAL EXPOSURE OBTAINED DURING INTERNSHIP

PROGRAMMING

Basics of Python programming

Control flow statements

File handling and scripting

Working with Google Colab

CYBERSECURITY CONCEPTS

Fundamentals of cybersecurity

Information security principles (CIA Triad)

History of cyber attacks

Offensive and defensive security concepts

SECURITY TOOLS & TECHNOLOGIES

Virtual cybersecurity lab setup

Linux command line basics

Network monitoring concepts

Vulnerability scanning tools

ADVANCED TOPICS

Basics of Generative AI &Large Language Models

Prompt engineering techniques

Open Source Intelligence (OSINT)

Intrusion Detection System (IDS)

Intrusion Prevention System (IPS)

Honeypot security concept

SKILLS LEARNED DURING INTERNSHIP

TECHNICAL SKILLS

Python programming basics | Understanding of cybersecurity fundamentals | Knowledge of CIA Triad | Linux command-line operations | Network traffic analysis | Vulnerability assessment concepts

Understanding cyber attack techniques

ANALYTICAL & PRACTICAL SKILLS

Network packet analysis | Reconnaissance and information gathering

Security monitoring and threat detection | Setting up and working in a virtual cybersecurity lab

SOFT SKILLS

Communication | Problem solving | Analytical thinking | Time management

Continuous learning through daily quizzes and assignments

TOOLS USED DURING INTERNSHIP

NETWORK ANALYSIS TOOLS

Wireshark – packet capture and network analysis

tcpdump – monitoring network traffic

PENETRATION TESTING & SCANNING

Metasploit – exploitation and vulnerability testing

Nmap – network discovery and port scanning

Nessus – vulnerability assessment

RECONNAISSANCE & INTELLIGENCE

Maltego – OSINT data analysis

Shodan – discovering internet-connected devices

Reconnaissance tools for information gathering

SECURITY FRAMEWORK

MITRE ATT&CK – understanding attacker techniques and tactics

CONCLUSION

ROOMAN
INNOVATE INTEGRATE EMPOWER

- The internship provided a **strong foundation in cybersecurity and programming.**
- Gained knowledge of **Python programming and Google Colab environment.**
- Understood **information security principles such as the CIA Triad.**
- Learned about **cyber attacks, vulnerability assessment, and security monitoring concepts.**
- Explored **AI concepts and prompt engineering related to cybersecurity.**
- Developed **practical skills through virtual cybersecurity labs.**
- Daily **quizzes were conducted to evaluate understanding of each topic.**